

Scalability & Future Plan

Date	7 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited support for concurrent users	System performance may degrade or freeze during traffic spikes when many students generate quizzes or request answers simultaneously	High
2	Limited integration with external educational sources	Restricts comprehensive curriculum mapping, leaving roadmaps reliant solely on direct AI generation without alignment to specific textbooks	High
3	No long-term cloud storage of user histories and recommendations	Users lose access to previous learning paths, customized roadmaps, or historical quiz performance scores once the local session expires	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a single server, optimized for personal Mac M1 environments	Deploy on cloud infrastructure (AWS or Google Cloud) with an ASGI load balancer and auto-scaling to distribute quiz processing traffic
2	Data Storage	Basic local storage using lightweight schemas for temporary profiles and recommendations	Implement scalable production databases like PostgreSQL or MongoDB to cleanly track complex student performance histories
3	Performance	AI tutoring responses and local text summaries are generated in real-time, risking local memory spikes	Use Redis caching mechanisms for common academic topics and run model queues asynchronously using Celery task workers
4	Security	Basic authentication patterns and standard internal API	Implement OAuth2, JWT role-based access control (Student vs. Educator), and secure API gateways to safeguard internal AI prompt

		structure setups	payloads
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration with GitHub Classroom, Anki Flashcards, and cloud-based file repositories.	2 Months	Enables automated student profile ingestion and direct synchronization of generated study decks.
Phase 3	AI-powered voice tutoring simulations, conversational essay grading, and automated study scheduling.	4 Months	Improves retention, simulates interactive verbal oral exams, and optimizes study block scheduling automatically.
Phase 4	Advanced visual analytics dashboard, localized multilingual translation support, and native mobile application ports.	6 Months	Expands global student accessibility, displays precise topic retention graphs, and enables offline learning anywhere.